

Decidable problems (recursive languages) about CFL

C1 $A_{CFG} = \{ \langle G \rangle \langle w \rangle : G \text{ is a CFG that generates } w \}$

M_{C1} decides A_{CFG}

C2 $A_{PDA} = \{ \langle P \rangle \langle w \rangle : P \text{ is a PDA that accepts } w \}$

M_{C2} = on input $\langle P \rangle \langle w \rangle$

1. convert P into an equivalent CFG G

2. run M_{C1} on $\langle G \rangle \langle w \rangle$

3. If M_{C1} accept $\langle G \rangle \langle w \rangle$,
accept $\langle P \rangle \langle w \rangle$

4. else

reject

C3 $\bar{E}_{CFG} = \{ \langle G \rangle : G \text{ is a CFG with } L(G) = \emptyset \}$

$S \Rightarrow^* u \text{ for } u \in \Sigma^*$?

M_{C3} = on input $\langle G \rangle$

1. mark all the terminals and ϵ

(A) $\rightarrow abc$

(B) $\rightarrow a(A)c$

$B \Rightarrow^* aabcc$

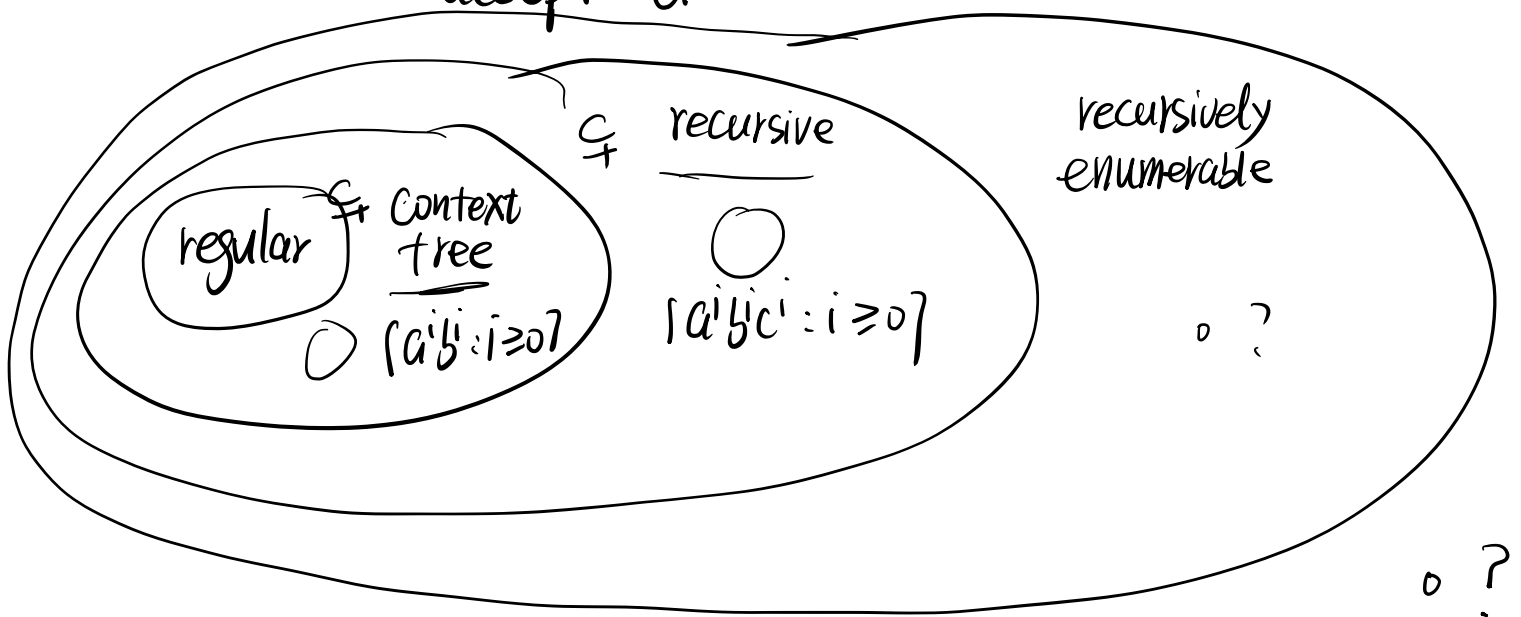
2. repeat until no new symbols are marked

If all the symbols on the RHS of a rule are marked
mark the symbol on the LHS

3. If S is mark,
reject " G "

4. else

accept " G "



1. $\text{recursive} \subseteq \text{recursive enumerable}$?
(R.E.)

2. $\exists L, L \notin \text{R.E.}$? yes

Let $f: A \rightarrow B$ be a function

f is injective if $f(a) \neq f(a')$ for any $a \neq a'$

f is surjective if $\forall b \in B, \exists a \in A, b = f(a)$

f is bijective if f is injective and surjective

Two sets A and B are equinumerous if $\exists$ bijection $f: A \rightarrow B$

A set is countable if it is finite or it is equinumerous with $\mathbb{N}$,
and uncountable otherwise

Theorem

Let A be a set.

The following statements are equivalent.

(a) A is countable

(b) $\exists$ injection $f: A \rightarrow \mathbb{N}$.

(c) there exists some way to enumerate the elements of A such that each element is listed within finite steps.

If A is finite, trivial.

Suppose A is infinite.

(a) $\Rightarrow$ (c) $\Rightarrow$ (b) $\Rightarrow$ (a)

(a) $\Rightarrow$ (c) $\exists$ bijection $f: A \rightarrow \mathbb{N}$.

list the elements of A in increasing order of $f(a)$

$\forall a \in A$. $f(a)$ steps.

(c) $\Rightarrow$ (b) for $a \in A$, $f(a) = \#$ steps used to list a for the first time.

(b) $\Rightarrow$ (a) $\exists$ injection $f: A \rightarrow \mathbb{N}$.

sort elements of A in increasing order of $f(a)$

$g(a) = \text{rank of } a$

$\Rightarrow$ bijection

Corollary

Any subset of a countable set is countable.

Proof:

A is countable

$\Rightarrow \exists$ injection $f: A \rightarrow \mathbb{N}$.

$A' \subseteq A \quad \hookrightarrow : A' \rightarrow \mathbb{N}$

A' is countable

Theorem

Let Σ be an alphabet. Σ^* is countable.

Proof:

If $\Sigma = \emptyset$, $\Sigma^* = \{\epsilon\}$ is countable.

Suppose $\Sigma \neq \emptyset$.

Enumerate all strings in Σ^* by increasing order of length.

A string of length i can be listed within

$|\Sigma|^0 + |\Sigma|^1 + \underbrace{|\Sigma|^2}_{\text{step}} + \dots + |\Sigma|^i$

$|\Sigma|^k = \#$ strings of length k

Σ^* is countable.

Corollary

The set of all Turing machines is countable.

Theorem

Let Σ be some non-empty alphabet.

Let $\mathcal{L}$ be the set of all languages over Σ .

$\mathcal{L}$ is uncountable

$\Rightarrow \exists L,$

L is not
R.E.

Proof:

Suppose that L is countable. X

The language in L can be listed as

$L_1, L_2, L_3, \dots$

Since Σ^* is countable, the strings in Σ^* can be listed as

$s_1, s_2, s_3, \dots$

$D = \{s_i : s_i \notin L_i\}$ is a language over Σ .

for any i , $s_i \in D$ iff $s_i \notin L_i$

$\Rightarrow D \neq L_i$

$\Rightarrow D \notin L$. $\longrightarrow$ contradiction

$\rightarrow$ diagonalization

Σ^*	s_1	s_2	s_3	$\dots$
L_1	1	0	0	$\dots$
L_2	0	0	0	$\dots$
L_3	1	1	0	$\dots$
$\dots$	$\dots$	$\dots$	$\dots$	$\dots$

$D = 0 \ 1 \ 1 \ \dots$

$D \neq L_i$

$D \notin L$.

recursive $\neq$ R.E. ? yes

$$H = \{ \langle M \rangle \langle w \rangle : M \text{ is a TM that halts on } w \}$$

- ✓ 1. H is R.E.
- ✓ 2. If H is recursive, then all R.E. languages are recursive.
- ✓ 3. H is not recursive.

Lemma

H is recursively enumerable (R.E.)

Proof:

$U =$ on input $\langle M \rangle \langle w \rangle$ universal TM
1. run M on w

$$\underline{U \text{ halts on } \langle M \rangle \langle w \rangle \Leftrightarrow M \text{ halts on } w \Leftrightarrow \langle M \rangle \langle w \rangle \in H}$$

U semidecides H .

Lemma

If H is recursive, so are all R.E. languages.

(H is complete for the class of R.E. languages)

Proof:

Let A be a R.E. language.

Let M be a TM that decides A .

Is $w \in A$? $\iff$ does M halt on w ? $\iff$ Is $"M" "w" \in H$?

If H is recursive, $\exists$ TM M_H decides H .

M_A = on input w

1. run M_H on $(M) "w"$

2. If M_H accept $"M" "w"$,
accept w

3. else
reject w .

Theorem

H is not recursive.

Proof:

$H = \{ "M" "w" : M \text{ is a TM that halts on } w \}$

$H' = \{ "M" : M \text{ is a TM that halts on } "M" \}$

H_d = $\{ "M" : M \text{ is a TM that does not halts on } "M" \}$

claim 1. If H is recursive, then H_d is recursive

Claim 2. H_d is not recursively enumerable



H is not recursive.

Proof of Claim 1

If H is recursive, $\exists M_H$ decides H .

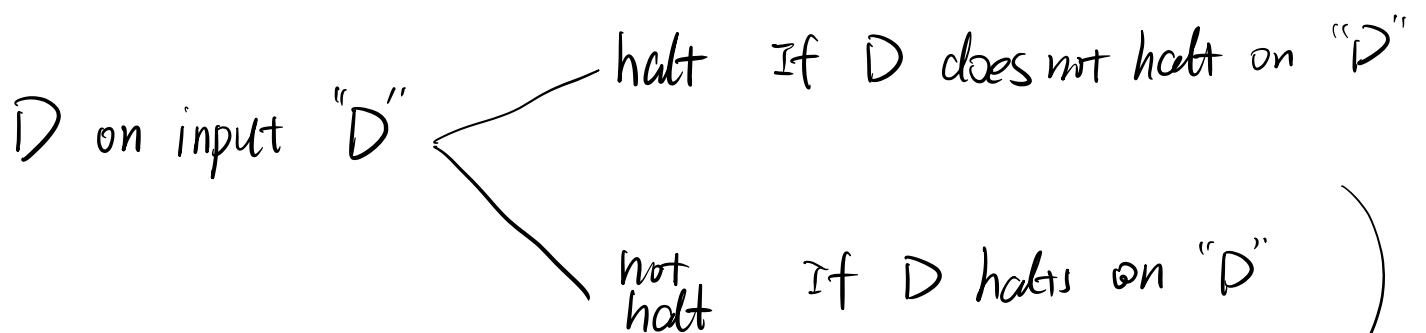
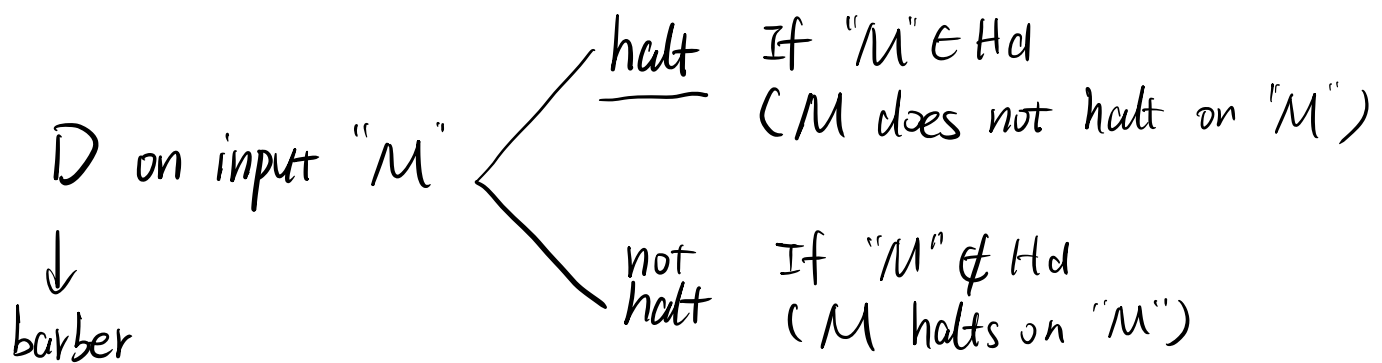
$M_d =$ on input " M "

1. run M_H on " M " " M "
2. If M_H accepts " M " " M ",
reject " M "
3. else
accept " M ".

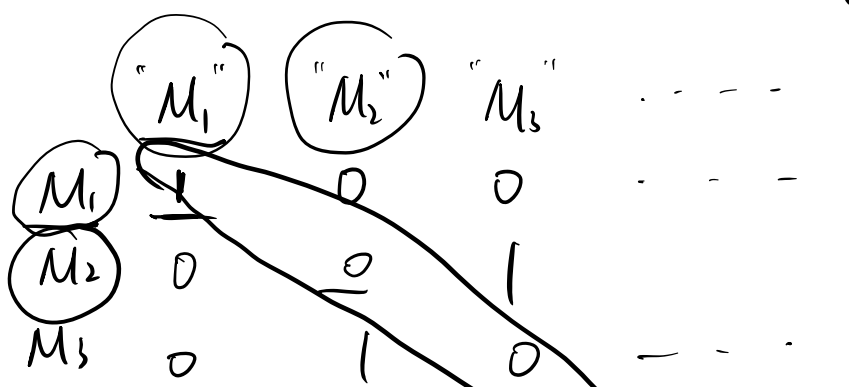
Proof of Claim 2 (H_d is not recursively enumerable)

Suppose H_d is R.E. $\times$

Some TM D semidecides H_d

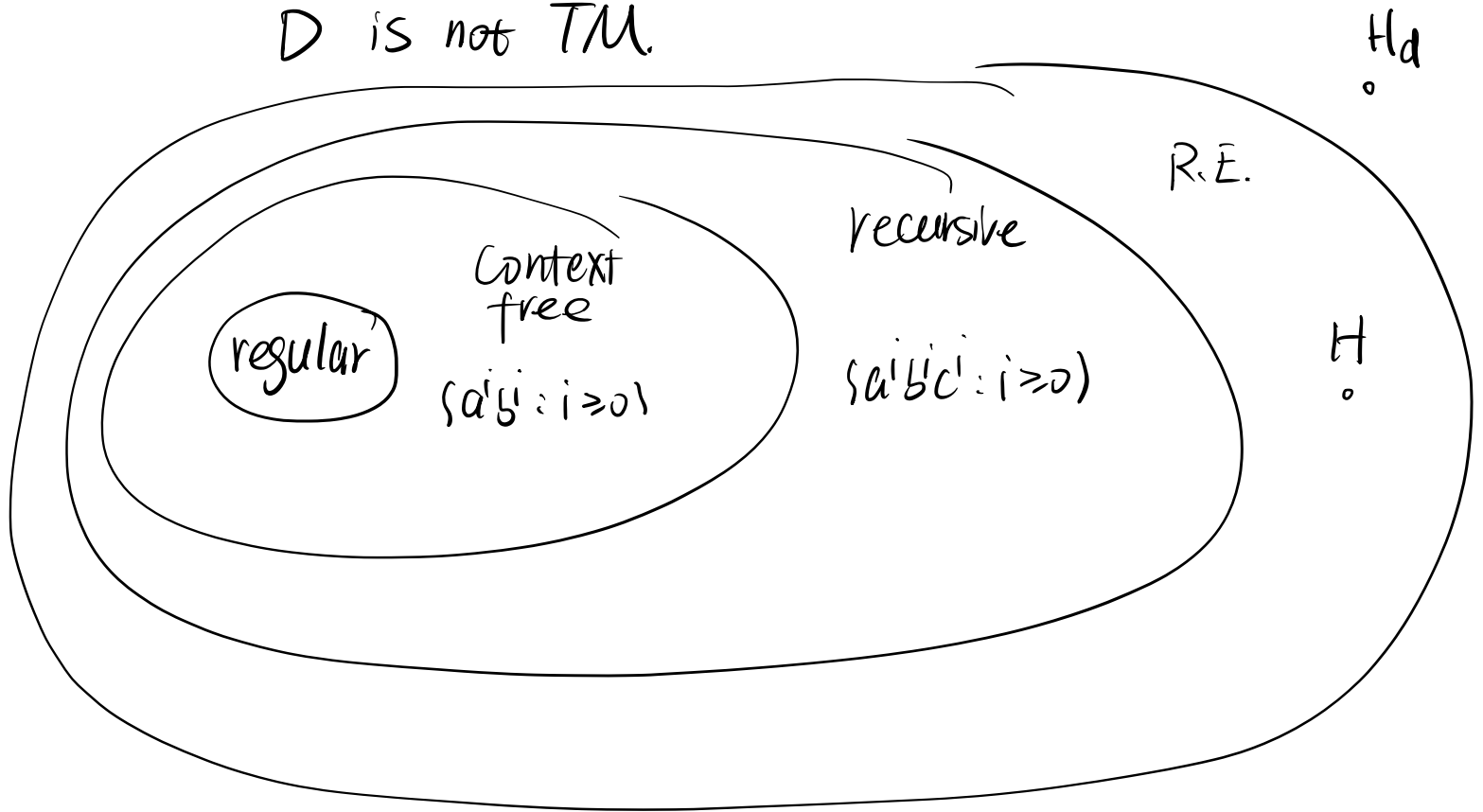


Contradiction



$D = 0 \quad 1 \quad 1 \quad \dots$

D is not TM.



Theorem

The class of R.E. languages is not closed under complement.

Proof.

H is R.E.

H is not recursive.

$\bar{H}$ is not R.E.

Lemma

L is recursive if and only if L and $\bar{L}$ are R.E.

Proof:

$\Rightarrow$) L is recursive $\Rightarrow \bar{L}$ is recursive $\Rightarrow \bar{L}$ and L are R.E

$\Leftarrow$) L and $\bar{L}$ are R.E

$\uparrow$ semi-decides $\uparrow$
 M_L $M_{\bar{L}}$

M^* = on input w

1. run M_L and $M_{\bar{L}}$ on w in parallel
2. If M_L halts,
accept w .
3. else // $M_{\bar{L}}$ halts
reject w .

Closure properties about recursive languages and R.E.

	recursive	R.E
$\cup$	$\checkmark$	$\checkmark$
$\cap$	$\checkmark$	$\times$
$-$	$\checkmark$	$\times$
$\circ$	$\checkmark$	$\checkmark$
$*$	$\checkmark$	$\checkmark$

If A and B are recursive, so is $A \cup B$.

$\uparrow$ decides $\uparrow$
 M_A M_B

$\uparrow$
 M_U

$M_U =$ on input " w "

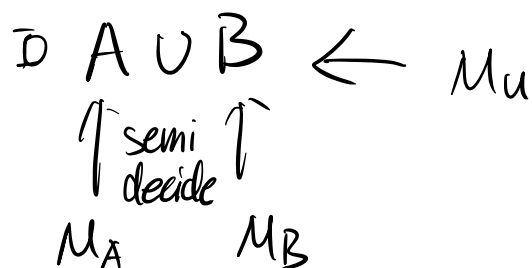
1. run M_A and M_B on " w "
2. If at least one of them accept w accept.
3. else reject

If A and B are recursive. so is $A \circ B$



$M_U =$ on input w $\rightarrow$ finite.

1. enumerate all possible pairs (u, v) s.t $uv = w$
 2. run M_A on u and run M_B on v
- - - - -



$M_U =$ on input w

1. run M_A and M_B on w
2. If one of them halt, halt.

M_U halts on w iff $w \in A$ or $w \in B$

If A and B are R.E., so is $A \circ B$



M_A M_B

NTM M_0

M_0 = on input " w "

1. non-deterministically divide $w = uv$
2. run M_A on u and run M_B on v
3. If M_A and M_B halt
halt.